Joe Lopez

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WORK EXPERIENCE

Microsoft Corporation — **Redmond, WA**

FIRMWARE DEVELOPMENT ENGINEER — **SEP 2021 – JUL 2023**

- Developed UEFI firmware for PCIe hardware intialization on company's first purpose-built ARM server platform, Cobalt 100. Ported open source libraries and contributed fixes upstream.
- Mitigated boot issues during chip and platform bringup by diagnosing ACPI issues using Windows Debugger (WinDbg) and ACPI tools, releasing a firmware patch in a timely manner to unblock others.
- Led the integration of firmware unit test suite into platform codebase, consuming open source framework and modifying it for our usecase. Worked with internal open-source owners to upstream enhancements for consumption by all first-party and third-party UEFI teams.
- Integrated pipeline for unit test CI/CD into platform codebase, using code coverage tools to compare coverage statistics against old version.
- Enhanced platform Reliability, Availability, and Serviceability (RAS) by constructing drivers for PCIe error reporting and handling, including workarounds for various silicon errata.

SOFTWARE DEVELOPMENT ENGINEER 2 — **JUL 2023 – MAY 2025**

- Examined codebase of first-generation product to plan and implement second-gen firmware based on new hardware layout. Developed design plan and distributed work amongst vendor team and oversaw firmware delivery timeline.
- Implemented novel features such as PCIe hot plug according to platform, OS, and protocol specifications. Ensured that no regressions were merged by creating integration tests to exercise different firmware and OS interactions with no human intervention necessary.
- Leveraged internally developed tool and agentic AI to generate firmware unit tests for previously uncovered lines of code using techniques such as RAG and multi-model approaches to achieve higher efficiency and scale to larger code files.
- Induced cross-team collaboration by facilitating work forum for PCIe/CXL hardware and software stakeholders for design reviews, bug investigations, and tech talks.

Robert Bosch Tool Corporation (Dremel) — **Mount Prospect, IL**

ELECTRICAL ENGINEERING INTERN — **MAY 2019 – AUG 2019**

- Collaborated with Electrical and Mechanical engineers to create prototype tool with USB (UART) interface to decrease reliance on bluetooth chips
- Improved data acquisition capacity by rewriting firmware for custom tool logger
- Assembled and diagnosed issues on prototype tools using oscilloscope, multimeter

EDUCATION

University of Illinois — **Urbana-Champaign, IL**

BACHELOR'S IN COMPUTER ENGINEERING — **AUG 2017 – JUN 2021**

- Relevant coursework in Systems Programming, Computer Science, Silicon

LEADERSHIP EXPERIENCE

Illini Formula Electric

DATA ACQUISITION LEAD — **AUG 2019 – AUG 2020**

- Revised effectiveness of the previous data acquisition system in order to design a system that logs vehicle data with LORA radio, GPS, Wi-Fi, and serial communication.
- Optimized costs and performance by designing multiple custom PCB's with Altium and implemented logging firmware/software between microcontroller and Raspberry Pi.

Theta Tau Professional Engineering Fraternity

PROJECTS COMMITTEE CHAIR — **AUG 2019 – JUN 2020**

- Managed a committee of 18 members to develop a project for presentation
- Introduced new technologies to the fraternity by hosting a microcontroller workshop

SKILLS

Programming Languages: C, C++, UEFI Framework, Python, Azure CI/CD, System Verilog, x86
Embedded Electronics: ARM, PCIe, STM32, I2C, SPI, UART, CAN
Miscellaneous: Altium Designer, Solidworks, GDB, Git, development in Windows/Linux environments